

ALU 4 BIT

The ALU performs 8 operations selected using a 3-bit select input.

1. Inputs and Outputs

Signal	Width	MSB	LSB	Purpose	
A	4-bit	A[3]	A[0]	First input	
B	4-bit	B[3]	B[0]	Second input	
sel	3-bit	sel[2]	sel[0]	Operation selection	
Y	4-bit	Y[3]	Y[0]	ALU result	
carry	1-bit	—	—	Carry/borrow indication	

MAIN CODE

```
module alu_4bit (  
    input wire [3:0] A,  
    input wire [3:0] B,  
    input wire [2:0] sel,  
    output reg [3:0] Y,  
    output reg carry  
);
```

```
always @(*)  
begin  
    // Default values  
    Y = 4'b0000;  
    carry = 1'b0;
```

```
    case (sel)
```

```
        3'b000: begin
```

```
    {carry, Y} = A + B;  
end
```

```
3'b001: begin  
    {carry, Y} = A - B;  
end
```

```
3'b010: begin  
    Y = A & B;  
end
```

```
3'b011: begin  
    Y = A | B;  
end
```

```
3'b100: begin  
    Y = A ^ B;  
end
```

```
3'b101: begin  
    Y = ~A;  
end
```

```
3'b110: begin  
    {carry, Y} = A + 1'b1;  
end
```

```
3'b111: begin  
    {carry, Y} = A - 1'b1;  
end
```

```
        default: begin
            Y = 4'b0000;
            carry = 1'b0;
        end

    endcase
end

endmodule
```

TEST BENCH CODE

```
`timescale 1ns/1ps
```

```
module alu_4bit_tb;
```

```
    reg [3:0] A;
    reg [3:0] B;
    reg [2:0] sel;
```

```
    wire [3:0] Y;
    wire carry;
```

```
// Instantiate ALU
```

```
alu_4bit uut (
    .A(A),
    .B(B),
    .sel(sel),
    .Y(Y),
    .carry(carry)
);
```

initial

begin

// Addition: $5 + 3 = 8$

A = 4'b0101;

B = 4'b0011;

sel = 3'b000;

#10;

// Subtraction: $5 - 3 = 2$

A = 4'b0101;

B = 4'b0011;

sel = 3'b001;

#10;

// AND: $5 \text{ AND } 3 = 1$

A = 4'b0101;

B = 4'b0011;

sel = 3'b010;

#10;

// OR: $5 \text{ OR } 3 = 7$

A = 4'b0101;

B = 4'b0011;

sel = 3'b011;

#10;

// XOR: $5 \text{ XOR } 3 = 6$

A = 4'b0101;

B = 4'b0011;

sel = 3'b100;

```
#10;
```

```
// NOT A
```

```
A = 4'b0101;
```

```
B = 4'b0000;
```

```
sel = 3'b101;
```

```
#10;
```

```
// Increment:  $5 + 1 = 6$ 
```

```
A = 4'b0101;
```

```
B = 4'b0000;
```

```
sel = 3'b110;
```

```
#10;
```

```
// Decrement:  $5 - 1 = 4$ 
```

```
A = 4'b0101;
```

```
B = 4'b0000;
```

```
sel = 3'b111;
```

```
#10;
```

```
$finish;
```

```
end
```

```
initial
```

```
begin
```

```
    $monitor("Time=%0t | A=%b | B=%b | SEL=%b | Y=%b | CARRY=%b",
```

```
        $time, A, B, sel, Y, carry);
```

```
end
```

```
endmodule
```